

The Invisible World

II PUC · Ch 8 · Microbes in Human Welfare

Starring Neha 👧 & Arjun 👦 — shrunk!

A BAIO storybook by Narendra Sir · Biology Lecturer



PAGE 1 OF 8

Shrinking Down

Neha and Arjun pressed the button on an old microscope — and shrank! They found themselves in a hidden world full of **bacteria**, **fungi**, **protozoa** and **viruses**, and even tinier **viroids** and **prions**. "These are **microbes**," boomed a voice. "Some cause disease, but most are helpers. Let us show you."

💡 REMEMBER

Microbes are present everywhere — soil, water, air, inside our bodies and even in hot springs and deep under the soil.



The Kitchen Helpers

In a bowl of milk, rod-shaped **Lactobacillus** were turning it into curd. "We are **lactic acid bacteria** (LAB). We also raise vitamin B_{12} in curd and fight harmful germs in your stomach." Nearby, **baker's yeast** — Saccharomyces cerevisiae — puffed up bread dough. In a cheese shop, **Propionibacterium sharmanii** blew the big holes in Swiss cheese.

💡 **REMEMBER**

The puffed-up appearance of dosa and idli dough is due to fermentation by bacteria.



The Factory Floor

At a factory, yeast made wine and beer without distillation, and whisky and rum with distillation. In a lab, they met **Alexander Fleming**'s famous mould — **Penicillium notatum** — which killed Staphylococci. "This was the first **antibiotic**, **penicillin**," said Arjun. Chain and Florey later proved how well it worked.

💡 **REMEMBER**

Acids from microbes: citric acid (Aspergillus niger), acetic acid (Acetobacter aceti), butyric acid (Clostridium butylicum), lactic acid (Lactobacillus).



Medicines and Enzymes

Bottles lined the shelves. "**Streptokinase** from Streptococcus is a clot buster for heart-attack patients," read Neha. "**Cyclosporin A** from Trichoderma polysporum stops organ rejection, and **statins** from the yeast Monascus purpureus lower blood cholesterol." **Lipases** in detergents remove oily stains; **pectinases** and proteases make bottled juices clear.

💡 **REMEMBER**

Bioactive molecules from microbes are used as medicines, enzymes and food additives.



Cleaning Dirty Water

They floated into a **sewage treatment plant**. In **primary treatment**, grit and floating solids were removed physically. In **secondary treatment**, air was pumped in and microbes grew into masses called **flocs**, eating up the organic matter and lowering the **BOD**. The settled **activated sludge** went into **anaerobic sludge digesters**.

💡 **REMEMBER**

BOD (biochemical oxygen demand) measures organic matter in water; the higher the BOD, the more polluted the water.



Gas from Cow Dung

In the sludge digesters and in a village gobar gas plant, the **methanogens** were at work — bacteria like **Methanobacterium** that also live in the rumen of cattle. They produced **biogas**: mainly methane with CO_2 and H_2. "It cooks food and lights homes," said a farmer. The technology was developed in India by IARI and KVIC.

💡 **REMEMBER**

Cow dung is rich in methanogens, so it is the main raw material of biogas plants.



Farmers' Tiny Friends

In a field, ladybirds ate aphids — **biocontrol**. Spores of **Bacillus thuringiensis** (Bt) killed caterpillars, the fungus **Trichoderma** protected roots, and **baculoviruses** attacked only insects. Underground, **Rhizobium** fixed nitrogen in legume root nodules, **mycorrhiza** fungi like Glomus helped absorb phosphorus, and cyanobacteria like **Anabaena** fed rice paddies. These are **biofertilisers**.

💡 **REMEMBER**

Free-living nitrogen fixers in soil: Azotobacter and Azospirillum.



Friends and Foes

The microscope beeped and the children grew back to normal size. "Not all microbes are friends," Neha said, remembering Salmonella that causes typhoid. "That is why we wash our hands — to remove harmful germs. And viruses are different from bacteria: they are not cells and can multiply only inside a host." Arjun nodded. "But without microbes, there would be no curd, no medicines — and no recycling of nutrients on Earth."

💡 **REMEMBER**

Decomposer microbes break down dead matter and return nutrients to the soil.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

bacteria p.1 · fungi p.1 · protozoa p.1 · viruses p.1 · viroids p.1 · prions p.1 · microbes p.1 · Lactobacillus p.2 · lactic acid bacteria p.2 · baker's yeast p.2 · Propionibacterium sharmanii p.2 · Alexander Fleming p.3 · Penicillium notatum p.3 · antibiotic p.3 · penicillin p.3 · Streptokinase p.4 · Cyclosporin A p.4 · statins p.4 · Lipases p.4 · pectinases p.4 · sewage treatment plant p.5 · primary treatment p.5 · secondary treatment p.5 · flocs p.5 · BOD p.5 · activated sludge p.5 · anaerobic sludge digesters p.5 · methanogens p.6 · Methanobacterium p.6 · biogas p.6 · biocontrol p.7 · Bacillus thuringiensis p.7 · Trichoderma p.7 · baculoviruses p.7 · Rhizobium p.7 · mycorrhiza p.7 · Anabaena p.7 · biofertilisers p.7



QUICK QUIZ

Do you remember?

1. Name one useful microorganism.

- Plasmodium
- Lactobacillus
- HIV
- Salmonella

2. How do bacteria differ from viruses?

- Bacteria are cells; viruses are not cells and need a host to multiply
- Viruses are bigger
- Bacteria cannot reproduce
- There is no difference

3. Why should we wash our hands?

- To make them dry
- To remove harmful germs
- To get vitamins
- To grow microbes

4. Penicillin was discovered by:

- Louis Pasteur
- Alexander Fleming
- Robert Koch
- Edward Jenner

5. Biogas is produced by:

- Lactobacillus
- Methanogens
- Rhizobium
- Yeast

Answers: 1-Lactobacillus · 2-Bacteria are cells; viruses are not cells and need a host to multiply · 3-To remove harmful germs · 4-Alexander Fleming · 5-Methanogens